

Nearest Neighbors

Project: Responsible Plant Classifier

1. Nearest Neighbors

Teach the learning cycle

Primary target: Compute absolute distances and rank training neighbours

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/22, ai-4/12. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf predictions from training examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one. Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Dataset Cards

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Label Quality

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Write an observable label rule with boundaries and a review option

Reference example, not the only acceptable wording: Touching is blocked. An object entirely outside is clear only when all objects are outside and the entire lane is visible. A partly hidden lane requires review because clear cannot be established from the available description.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training lengths 2(H),6(S),10(H); query=8. Compute distances. Explain the tie and why combining length in mm with width in cm changes a two-feature distance.

Concept to explain

k-nearest neighbors predicts from nearby labeled examples under a chosen feature scale, distance rule, and value of k.

Term · Meaning

Term: distance

Meaning: numeric measure of dissimilarity

Term: neighbor

Meaning: nearby labeled training example

Term: k

Meaning: number of neighbors voting

Term: scaling

Meaning: make feature ranges comparable

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Use a single numeric feature to make distance arithmetic visible, then distinguish neighbour selection from voting. Store only training rows. Alphabetical IDs settle equal distances for this classroom method; explain that it is a convention and other implementations can use different tie behavior.

Readiness answer

It stores a root question and leaf predictions learned from training. A new example follows the appropriate branch to a stored leaf prediction.

Printed lesson and workbook

Number line from 0 to 14 cm and five labelled training markers

Optional Python 3 editor with no packages required

End goal for the pictorial case

By the end, I can explain nearest neighbors using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Calculate distances and compare $k=1$ with $k=3$.

MODEL IDEA k -nearest neighbors predicts from nearby labeled examples under a chosen feature scale, distance rule, and value of k .

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD New leaf has normalized features (2, 3). Using Manhattan distance, add the absolute coordinate differences. Training leaves at (2, 2), (4, 3), and (1, 5) have distances 1, 2, and 3 from (2, 3).

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 New leaf has normalized features (2, 3).

STEP 2 Using Manhattan distance, add the absolute coordinate differences. Training leaves at (2, 2), (4, 3), and (1, 5) have distances 1, 2, and 3 from (2, 3).

STEP 3 With $k=1$, nearest label wins; with $k=3$, majority among all three wins.

STEP 4 Rescale features and retest because centimeters must not overwhelm a 0–1 color score.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

One-dimensional training points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute distance.

Find $k=1$ neighbours and handle ties.

Exact explanation to model

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. Scaling matters with differently measured features.

Explain the added worked example

A nearest-neighbour classifier predicts using nearby training examples rather than following a fitted tree. This week we store labelled examples, measure how different their feature values are from a new query, and let selected neighbours vote. Nearby means close under our chosen feature and distance, not necessarily similar in every real-world way.

Our new fictional dataset has a distance reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity. IDs identify rows and break distance ties; they are not numeric features. These invented readings are not a robot safety system.

For this one-dimensional feature, distance is `abs(training reading minus query reading)`. `abs` removes the sign. Set `k=3` and give each selected neighbour one vote. Order equal distances by ID alphabetically, declared before scoring. In this binary, three-vote exercise the label vote cannot tie, but the distances can.

For query 3, distances to A through E are 1,1,5,7,9 cm. sorted orders tuples by distance first, then the unique ID. `ranked[:3]` takes the first three rows. The output is ['A', 'B', 'C'] then clear: one blocked vote and two clear votes.

If `k` were 1 under the same distance tie rule, A would win and predict blocked. Changing `k` can change a prediction. We declared `k=3` for this exercise; do not choose `k` after seeing the desired answer and pretend the method was unchanged.

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three training rows. It does not validate those conditions. A larger `k` than the stored row count would not produce the promised number of votes. Keep unresolved review labels under a separate documented policy.

Key terms

Term · Meaning

Term: Distance

Meaning: A numerical measure of difference between selected feature values

Term: Neighbour

Meaning: A stored training example close under the chosen distance

Term: `k`

Meaning: The number of neighbours selected for a prediction

Term: Vote

Meaning: A contribution of a selected neighbour's label to the predicted class

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Three selected neighbours for query 3

Stored example · Feature distance · One label vote

Stored example: A at 2 cm

Feature distance: abs 2 minus 3 is 1

One label vote: blocked

Stored example: B at 4 cm

Feature distance: abs 4 minus 3 is 1

One label vote: clear

Stored example: C at 8 cm

Feature distance: abs 8 minus 3 is 5

One label vote: clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Store the training examples and the method settings. At prediction time, compare the query only with stored training examples. Do not add the query's reference label or other validation labels to the neighbour pool. No cutoff or tree is fitted here, but retaining training examples is still part of the learning method.

Why do A and B tie for distance to query 3? Which wins for $k=1$ under our rule, and why does $k=3$ predict differently? Does the ID rule establish which label is true?

A vote share of two out of three describes these neighbours. It is not a verified two-thirds probability that the new scene is clear. A prediction can be wrong even when the nearest examples agree.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Nearest Neighbors

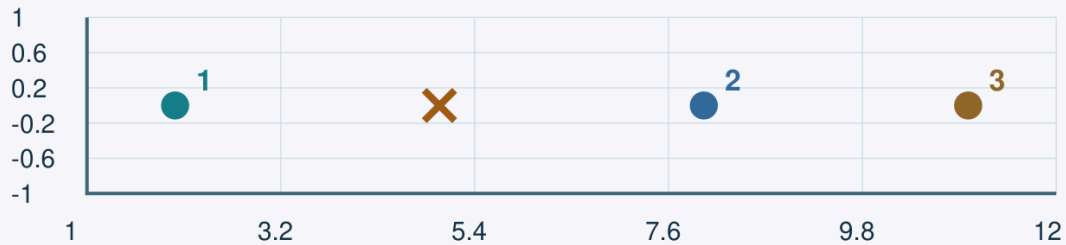
Let a nearest-neighbour picture show each distance before voting.

The facts we will use

Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. Distance is absolute difference; ties use alphabetically earlier ID.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, $y=0$

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Nearest Neighbors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

To A: $|5-2|=3$; B: $|5-8|=3$; C: $|5-11|=6$ cm.



2. Apply k=1 rule

A and B tie nearest; declared ID tie-break chooses A, predicting healthy.



3. Compare k=3

All three vote: two spotted and one healthy, so k=3 predicts spotted.

CHANGE ONE THING

Move the query to length 7 cm.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Measure distances

What happens and why: To A: $|5-2|=3$; B: $|5-8|=3$; C: $|5-11|=6$ cm.

Step: 2. Apply k=1 rule

What happens and why: A and B tie nearest; declared ID tie-break chooses A, predicting healthy.

Step: 3. Compare $k=3$

What happens and why: All three vote: two spotted and one healthy, so $k=3$ predicts spotted.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. All three vote: two spotted and one healthy, so $k=3$ predicts spotted. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Length alone is a toy feature, not a disease diagnosis. Distance choices and feature scaling can change neighbours.

Change one thing in the pictured case

Change: Move the query to length 7 cm.

Prediction: $k=1$ predicts spotted from B, distance 1; $k=3$ remains spotted.

Reason: Distances become 5, 1, 4 cm, so the nearest tie disappears.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: One-dimensional training points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute distance.

Write the question to explain: Find $k=1$ neighbours and handle ties.

Reveal after predictions: Distances 4, 2, 2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. Scaling matters with differently measured features.

Keep for an independent attempt: For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-neighbor with absolute distance, predict its label.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: $\text{abs}(2-3)$ and $\text{abs}(4-3)$ both equal 1. For $k=1$ alphabetical tie order chooses A, blocked. $k=3$ adds B and C, whose clear votes win 2 to 1. ID order is a deterministic convention, not evidence of the query's true label. Exit: using labelled validation examples in the neighbour pool leaks evaluation answers and may let a query select itself. Restore the original training-only pool and score every validation query without adding its label.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve tree storage and prediction. Introduce prediction from stored examples with a declared distance.

Stage: 7 to 18 minutes

Teacher action: Place A through E on the number line. Calculate absolute distances to query 3 and complete W1.

Stage: 18 to 32 minutes

Teacher action: Trace tuple sorting and list slicing. Complete W2; distinguish tied distances from a tied label vote.

Stage: 32 to 45 minutes

Teacher action: Read the diagram and answer the guided question. Explain why ID order is not evidence of truth.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the training-only neighbour pool. Ask why validation labels must remain outside it.

Stage: 8 to 20 minutes

Teacher action: Compute the three validation predictions and complete W3. Compare the fixed training-majority baseline.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before support. Require ordered IDs, both k predictions and the full validation score.

Stage: 35 to 45 minutes

Teacher action: Use W5 and the exit question. Offer the fresh retry after feedback and record support.

Reduce help deliberately

Completion task: For query 3, list the selected IDs and predictions for $k=1$ and $k=3$ using alphabetical ID order on equal distances. Explain what changes and what remains the same.

Teacher check: For $k=1$ select A and predict blocked. For $k=3$ select A,B,C and predict clear from two clear votes. Training data, distance and ID tie rule stay the same; the number of voting neighbours changes.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Nearest neighbours and distances. Proposed explanation: “A negative difference is a negative distance”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Take the absolute difference; in this example distance is nonnegative regardless of which reading is larger.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why is the $k=1$ outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

A healthy and B spotted are equally near. The declared alphabetical rule resolves the otherwise ambiguous one-neighbour selection. Changed-case check: $k=1$ predicts spotted from B, distance1; $k=3$ remains spotted. Distances become 5, 1, 4 cm, so the nearest tie disappears.

Changed-case reasoning: $k=1$ predicts spotted from B, distance1; $k=3$ remains spotted. Distances become 5, 1, 4 cm, so the nearest tie disappears.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Neighbor Map: compute distances, rank neighbors, compare k values, and show how scaling changes one result.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Nearest neighbor means the photo that looks most similar to a person.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A negative difference is a negative distance: Take the absolute difference; in this example distance is nonnegative regardless of which reading is larger.

Odd k prevents every kind of tie: Three binary votes avoid a label tie, but neighbours can still tie in distance, including at the selection boundary.

An exact match proves the label: It matches the selected feature only. Other conditions, annotation errors or contradictory examples can remain.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. Scaling matters with differently measured features.

Independent question

For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-neighbor with absolute distance, predict its label.

Expected independent answer

Distances are 4 to fern and 2 to grass, so predict grass. This result assumes the specified feature and distance rule.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Nearest-neighbor prediction depends on the order examples are listed.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Distances are 6,2,2, producing conflicting nearest labels. Use a predefined tie policy or abstain. Raw numeric distances depend on units; standardise compatible measurements and fit any learned scaling using training data only.

Explanations for the additional workbook examples

W1 Calculate feature distances

For query 3 cm, calculate distances to all five training rows. Explain why A and B tie even though their labels differ.

Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

W2 Apply neighbour and tie rules

For query 3, list the selected IDs and predictions for $k=1$ and $k=3$ using alphabetical ID order on equal distances. Explain what changes and what remains the same.

For $k=1$ select A and predict blocked. For $k=3$ select A,B,C and predict clear from two clear votes. Training data, distance and ID tie rule stay the same; the number of voting neighbours changes.

W3 Keep evaluation examples separate

Why must the three validation reference labels not enter the neighbour pool? Give the fixed $k=3$ score and baseline score on the worked validation examples.

Adding validation labels uses evaluation answers to make predictions and changes the intended comparison with unseen trips. The fixed $k=3$ method predicts clear three times and gets $1/3$. The training-majority clear baseline also gets $1/3$.

W4 Find neighbours independently

New training rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all distances, order the IDs by distance then alphabetically, and predict with $k=1$ and $k=3$. Using the same training pool and $k=3$, score validation (6,blocked),(10,clear),(12,blocked). Compare the training-majority baseline.

Query 6 distances P,Q,R,S,T are 5,1,1,5,7. Order Q,R,P,S,T. $k=1$ predicts blocked from Q; $k=3$ uses Q,R,P and also predicts blocked. For queries 6,10,12, selected triples are Q R P, S R T, S T R. All predict blocked, matching validation rows 1 and 3: $2/3$. Training-majority blocked also scores $2/3$.

W5 Interpret a vote and a match

Two of three neighbours vote clear. Does that prove a two-thirds chance of a correct prediction? Does an exact feature match guarantee a correct label? Explain both limits.

The vote fraction describes the selected training labels, not a verified probability of correctness. An exact match on one feature can still differ in omitted conditions or have a wrong or conflicting label. Separate evaluation and relevant features remain necessary.

Assess independent transfer

Selected case: Training points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute distance; for a nearest-neighbor tie choose the alphabetically earlier ID. Compare $k=1$ and $k=3$.

Selected case answer

Distances A6,B2,C2. The specified tie rule selects B for $k=1$, predicting spotted. With $k=3$, two healthy labels outvote one spotted. State the distance and tie policy; this tiny case establishes no wider accuracy.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Implement a transparent k -NN paper prototype for the Plant Classifier and record the chosen features,

scaling, distance, and k.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.
Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 13: Train a Classifier. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use physical steps on a number line before abs. Supply an empty distance table without the order.
Extension: explain how renaming tied IDs can change a prediction under this policy while the measured features remain unchanged, and why the policy must be recorded.

Require W4 distances 5,1,1,5,7, order Q,R,P,S,T, both predictions blocked, and validation plus baseline scores 2/3. Fresh retry: L at 0 blocked, M at 4 clear, N at 6 clear, O at 10 blocked, P at 12 clear; query 5. Distances are 5,1,1,5,7; order M,N,L,O,P. k=1 is clear from M; k=3 is clear from M,N,L. Ask why L wins the third-place distance tie over O: alphabetical ID policy. Require a training-only pool and no probability claim.

Week 11 covers scaling without leakage. Carry forward training-only storage and explicit distance choices. Teacher reference: <https://scikit-learn.org/stable/modules/neighbors.html> (nearest-neighbour classification and tie considerations; reviewed 2026-09-27).

Plan the delayed return

Suggested return points in this level: ai-4/13, ai-4/15, ai-4/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Compute absolute distances and rank training neighbours

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For query 3, list the selected IDs and predictions for $k=1$ and $k=3$ using alphabetical ID order on equal distances. Explain what changes and what remains the same.

Reasoned practice answer: For $k=1$ select A and predict blocked. For $k=3$ select A,B,C and predict clear from two clear votes. Training data, distance and ID tie rule stay the same; the number of voting neighbours changes.

Check every learner before discussion: Use the worked case for Nearest neighbours and distances.

Proposed explanation: “A negative difference is a negative distance”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Take the absolute difference; in this example distance is nonnegative regardless of which reading is larger.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Move the query to length 7 cm. Require a prediction and a reason before comparing. Expected result: $k=1$ predicts spotted from B, distance 1; $k=3$ remains spotted. Reason: Distances become 5, 1, 4 cm, so the nearest tie disappears.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.